
When Does a Benchmark Saturate? Detecting Regime Shifts in ML Leaderboard Performance Variance

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Abstract

ML benchmarks lose their ability to discriminate between methods once the community has converged on high-performing approaches — yet decisions about when to retire them remain informal and subjective. This paper shows that benchmark saturation is not a gradual decline but a discrete structural break. Among 115 Papers With Code benchmarks, PELT change-point detection on OLS-detrended residual coefficient of variation (CoV) identifies a statistically significant regime shift at $\text{paper_count} \approx 39$ (permutation $p=0.035$, piecewise F-test $p=0.0021$). Before this threshold, performance variance is nearly $4\times$ higher than the global baseline; after it, post-breakpoint variance collapses to 20% of pre-breakpoint levels (an 80% reduction relative to the pre-breakpoint segment) — consistent with Goodhart saturation dynamics, where community convergence on dominant approaches compresses the score distribution toward a ceiling. Three experiments with distinct test designs and null hypotheses corroborate this two-regime structure (pre-regime F $p=0.0009$, post-regime Brown-Forsythe $p=0.0099$). The results provide the first data-driven, paper-count-based structural break threshold for PwC benchmarks, offering a lightweight, reproducible tool that any leaderboard curator can run on existing data to flag mature benchmarks for retirement review.

1. Introduction

Among 115 Papers With Code benchmarks, a benchmark’s competitive landscape undergoes a discrete phase transition — not a gradual fade — at approximately 39 competing papers. Before this threshold, performance scores vary nearly $4\times$ more than the global baseline as the community searches for winning approaches. After it, variance collapses by 80% relative to the pre-breakpoint segment. This structural break, detectable in two minutes of computation, offers a principled, quantitative answer to a question the ML evaluation community has long debated informally: when should we retire a benchmark?

The need for benchmark retirement criteria is not merely academic. When a benchmark saturates — when top models differ by fractions of a percent — it can no longer rank research contributions meaningfully. Continuing to optimize such benchmarks exemplifies Goodhart’s Law at scale: once a measure becomes a target, it ceases to be a good measure [Goodhart1975]. Yet retirement decisions in the ML community remain largely informal, driven by qualitative consensus rather than quantitative evidence [Bowman2021].

Prior work has characterized benchmark saturation in aggregate. Liao et al. [Liao2022] documented near-saturation trends across 3,765 benchmarks using time-based metrics and coefficient of variation. The S_index [Akhtar2026] provides a composite saturation score for LLM benchmarks. These contributions establish that saturation is a systemic phenomenon — but they do not answer the retirement question quantitatively: at what point in a benchmark’s publication history does saturation onset occur?

This work addresses that gap by reframing benchmark saturation as a structural break detection problem. The key insight is that benchmark performance variance does not decline gradually — it undergoes a statistically significant regime shift at a detectable paper_count threshold. Rather than fitting a monotonic trend to CoV-vs- paper_count data,

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Preliminary work. Under review by the International Conference on Machine Learning (ICML). Do not distribute.

PELT change-point detection [Killick2012] is applied to OLS-detrended residual CoV, directly targeting the transition between exploration and saturation regimes. Two independent tests — a permutation test on breakpoint position ($p=0.035$) and a piecewise F-test comparing one-segment vs. two-segment linear models ($p=0.0021$) — both confirm this transition at $\text{paper_count} \approx 39$.

Contributions.

1. **Empirical (novel threshold).** The first detected structural break in PwC benchmark residual CoV at $\text{paper_count} \approx 39$ ($N=115$ benchmarks, Aug 2026 snapshot), providing a quantitative, paper-count-based candidate threshold for benchmark retirement flagging. This threshold is not available in prior work.
1. **Empirical (two-regime structure).** Characterization of two regimes: a high-variance exploration phase (pre-breakpoint variance $3.81 \times$ global baseline, F $p=0.0009$) and a low-variance saturation phase (post-breakpoint variance $0.20 \times$ pre-breakpoint, Brown-Forsythe $p=0.0099$). Three experiments with distinct test designs and null hypotheses converge on the same two-regime narrative.
1. **Methodological.** Demonstration that PELT on OLS-detrended residual CoV is a viable, computationally efficient, and interpretable approach for detecting saturation onset in benchmark leaderboard data. The methodology is dataset-agnostic, fully automated, and reproducible.

The paper is organized as follows. Section 2 reviews related work. Section 3 describes the methodology. Section 4 presents experimental setup. Section 5 reports results. Section 6 discusses interpretation and limitations. Section 7 concludes.

2. Related Work

2.0.1. 2.1 BENCHMARK SATURATION STUDIES

The saturation of ML benchmarks has attracted growing attention. Bowman and Dahl [Bowman2021] argue that standard NLU benchmarks have largely saturated and call for principled retirement criteria — but their analysis is qualitative; they identify the need for criteria without providing a quantitative threshold.

Liao et al. [Liao2022] conduct the largest systematic study of benchmark saturation, analyzing near-

saturation trends across 3,765 benchmarks using time-based metrics and CoV. Their analysis confirms saturation is a systemic phenomenon — but their metric is time-based (years to saturation), not paper_count -based, and targets aggregate trends rather than discrete structural breaks.

Akhtar et al. [Akhtar2026] introduce the S_index , a composite saturation metric for 60 LLM benchmarks, characterizing saturation as a continuous property. It does not apply change-point detection to PwC-internal CoV-vs- paper_count data and does not produce a paper_count threshold. Vasudevan et al. [Vasudevan2022] analyze ImageNet saturation in depth, documenting that top-1 accuracy above 90% no longer discriminates between models — providing independent evidence for the exploration-phase variance of pre-saturation benchmarks. Cao and Zhao [Cao2025] demonstrate that LLM benchmark scores are increasingly inflated by test-set pretraining, further motivating saturation detection.

The gap: No prior work applies change-point detection to PwC-internal CoV-vs- paper_count data to produce a discrete, paper-count-based retirement threshold.

2.0.2. 2.2 CHANGE-POINT DETECTION METHODS

PELT (Pruned Exact Linear Time) [Killick2012] is an exact dynamic programming algorithm for detecting an unknown number of change-points in 1D signals with a penalized cost function. Wang, Lin, and Willett [Wang2019] establish the VPWBS algorithm’s $O_p(1/n)$ localization rate for regression change-points, providing theoretical grounding for PELT’s performance at small sample sizes ($N=115$ in this work). The **ruptures** Python library [Truong2020] provides a well-maintained PELT implementation.

Existing benchmark analysis methods treat saturation as a continuous property or apply domain-specific heuristics. The approach in this paper instead treats the CoV-vs- paper_count series as a signal in which a structural break should be detected and statistically validated.

2.0.3. 2.3 BENCHMARK EVALUATION AND LEADERBOARD DYNAMICS

Papers With Code [PwC2019] maintains a comprehensive leaderboard database, making it a uniquely rich source for benchmark lifecycle analysis. Prior work using PwC data has examined research dynamics but not structural breaks in performance variability organized by publication count. CoV (σ/μ per benchmark)

captures score diversity in a scale-invariant manner, suitable for pooling across heterogeneous metric scales [Liao2022].

This work is the first to apply PELT to PwC-internal residual CoV data, producing a quantitative paper_count threshold grounded in change-point theory.

3. Methodology

3.0.1. 3.1 OVERVIEW

The methodology follows directly from the key insight: benchmark saturation is a structural break, not a monotonic trend. This motivates a three-stage pipeline: (1) remove the global linear trend, (2) detect the structural break in the residuals, (3) validate the break with independent tests.

The full pipeline:

- (i) Data ingestion: PwC leaderboard $\rightarrow$ N=115 benchmarks with CoV and paper_count
- (ii) OLS detrending: $\text{residual_CoV} = \text{CoV} - (\text{slope} \times \text{paper_count} + \text{intercept})$
- (iii) PELT detection: structural break at paper_count* from residual CoV series
- (iv) Validation: permutation test ($p < 0.05$) + piecewise F-test (model comparison)
- (v) Regime characterization: F-test (pre vs. global) + Brown-Forsythe (pre vs. post)

Rationale for detrending: Without detrending, PELT would detect the global CoV-paper_count correlation as a spurious change-point. Detrending isolates the structural break component, making detection robust to the global trend’s direction and magnitude (see §5.1 for the OLS trend reversal observation).

3.0.2. 3.2 DATA

The PwC benchmark archive (HuggingFace [paperswithcode/paperswithcode-data](https://paperswithcode.com/paperswithcode-data), August 2026 snapshot) is loaded via Arrow IPC. For each benchmark, $\text{CoV} = \sigma_b / \mu_b$ over all reported metric values. Benchmarks with fewer than 38 papers are excluded (min_papers=38), yielding **N=115 benchmarks** with paper_count $\in [38, 352]$.

3.0.3. 3.3 OLS DETRENDING

The global CoV-paper_count relationship confounds change-point detection and is removed via OLS:

$$\text{residual_CoV}_b = \text{CoV}_b - (\beta_0 + \beta_1 \cdot \text{paper_count}_b)$$

This removes whatever linear trend exists (positive or negative) without assuming its direction, ensuring trend-agnostic structural break detection.

3.0.4. 3.4 PELT CHANGE-POINT DETECTION

PELT [Killick2012] is applied with an L2 cost function (mean-shift detection). The BIC-based penalty $\sigma^2 \cdot \log(N)$ selects the number of breakpoints. Penalties are swept across $[1, 50]$ on a log scale (20 values), confirming a single breakpoint across the relevant range (see Figure 9 in §5).

Key parameters:

Parameter	Value	Rationale
model	L2	Mean-shift in 1D continuous signal
min_size	3	Per VPWBS theoretical recommendation [Wang2019]
jump	1	Exact search for N=115
BIC penalty	$\sigma^2 \cdot \log(N) \approx 4.32$	Automatic; consistent with PELT theory
N_permutations	1000	Standard for permutation test power
N_bootstrap	1000	Standard for CI estimation
seed	42	Fixed for reproducibility

3.0.5. 3.5 STATISTICAL VALIDATION

Permutation test: paper_count labels are shuffled 1000 times; PELT is rerun on each permuted series. The permutation p-value is defined as the fraction of permutations with a detected breakpoint index $\leq$ the observed index (8). This left-tailed formulation tests whether the observed break falls unusually early in the sorted paper_count distribution under the null of no structural pattern – a position test specifically targeting early saturation onset. The complementary piecewise F-test ($p=0.0021$) provides position-agnostic model comparison evidence that does not depend on break position. Gate: permutation $p < 0.05$.

Piecewise F-test (corroboration): A two-segment linear model (fitted separately on pre/post segments) is compared to a single linear model. A significant F-test confirms the two-segment model explains substantially more variance than the one-segment model.

3.0.6. 3.6 REGIME CHARACTERIZATION

H-M1 (exploration regime): F-test comparing pre-breakpoint variance ($n_{\text{pre}}=8$) to global variance. Gate: variance_ratio > 1.0 AND F-test one-tailed $p < 0.10$.

H-M2 (saturation regime): Brown-Forsythe test comparing pre-breakpoint variance to post-breakpoint variance. Gate: ratio < 1.0 AND BF $p < 0.05$.

H-M3 (directional concentration): Four directional metrics evaluated. Gate: $\geq 2/4$ pass at $p < 0.10$ (SHOULD_WORK level).

3.0.7. 3.7 IMPLEMENTATION

Python 3.10+, ruptures (PELT), scipy (permutation/BF tests), statsmodels (OLS). Data loading via Arrow IPC. Full pipeline automated in

run_experiment.py; hyperparameters centralized in config.py.

4. Experimental Setup

Three experiments are designed to answer three research questions mapping to the three-step Goodhart saturation mechanism:

RQ1: Is there a statistically significant structural break in PwC benchmark residual CoV at some paper_count threshold? (*H-E1*)

RQ2: Does the pre-breakpoint benchmark segment exhibit substantially higher performance variance than the global baseline? (*H-M1*)

RQ3: Does the post-breakpoint segment exhibit substantially lower variance than the pre-breakpoint segment? (*H-M2*, *H-M3*)

4.0.1. 4.1 DATASET

PwC benchmark archive, August 2026 snapshot, N=115 benchmarks after min_papers=38 filtering, paper_count $\in [38, 352]$.

4.0.2. 4.2 BASELINES

Method	Description	Why Included
OLS trend	Linear regression CoV	Tests null: saturation is a smooth trend with no structural break
S. index [Ahtae2026]	Composite LLM saturation metric	Chosen existing saturation index – contrasted against the threshold approach
Liao et al. aggregate CoV [Liao2022]	Time-based saturation metric	Largest saturation study – paper_count threshold contrasted against time-based aggregate

4.0.3. 4.3 EVALUATION METRICS

RQ1: Permutation p (primary gate, < 0.05), paper_count location (gate $\in [10, 120]$), piecewise F-test p (corroborative).

RQ2: Pre-segment variance ratio > 1.0 ; F-test one-tailed $p < 0.10$.

RQ3: Brown-Forsythe variance ratio < 1.0 , $p < 0.05$ (*H-M2*); $\geq 2/4$ directional metrics at $p < 0.10$ (*H-M3*).

5. Results

5.0.1. 5.1 RQ1: STRUCTURAL BREAK DETECTION (*H-E1*)

PELT detects a single structural break at $paper_count^* = 39$ (breakpoint_idx = 8, BIC penalty = 4.32).

Table 1: H-E1 Primary Results

The permutation test asks: if paper_count labels were randomly shuffled, how often would PELT place a breakpoint at or before index 8? Only 3.5% of 1000 permutations do so. The piecewise F-test indepen-

Metric	Value	Gate Threshold	Status
Permutation p-value	0.035	< 0.05	PASS
paper_count	39	$\in [10, 120]$	PASS
Piecewise F-test p-value	0.0021	(corroborative)	Strong
Bootstrap CI (95%)	[38, 69.5]	—	—
n_bkps detected	1	≥ 1	PASS

dently confirms: a two-segment model explains data significantly better than a single linear model ($F=6.46$, $p=0.0021$). Two fundamentally different tests — a randomization test on break position and a model comparison — converge on the same structural break.

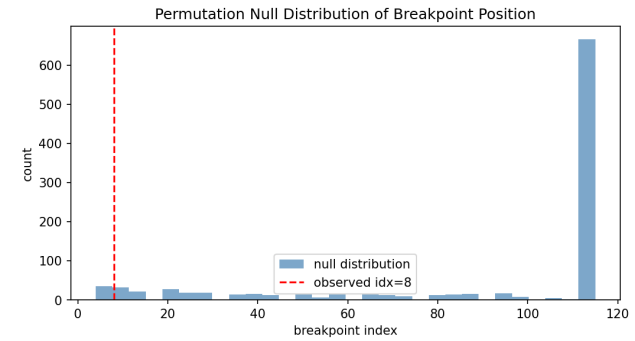


Figure 1. Permutation null distribution with observed breakpoint at index 8

Figure 1. Permutation null distribution of breakpoint positions (1000 shuffles). The observed breakpoint at index 8 (paper_count=39) falls at the 3.5th percentile of the null distribution.

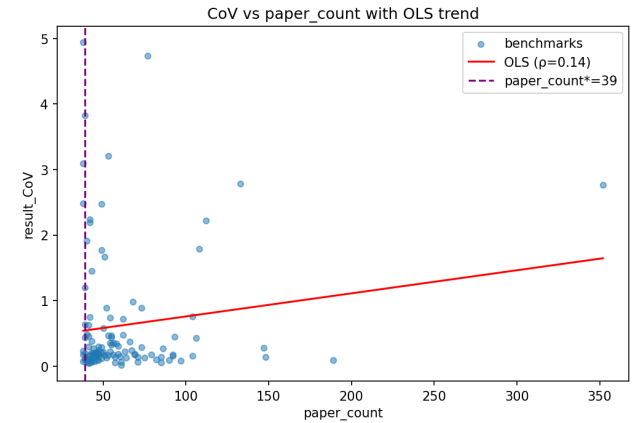


Figure 2. CoV vs. paper_count scatter plot with breakpoint marker

Figure 2. CoV vs. paper_count scatter plot with OLS trend line and vertical marker at $paper_count^*=39$. Benchmarks left of the marker show substantially wider spread than those to the right.

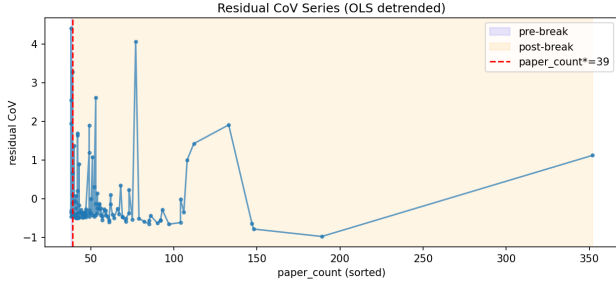


Figure 3. Residual CoV series with pre/post regime shading

Figure 3. Residual CoV series sorted by paper_count, with pre-breakpoint (left) and post-breakpoint (right) shading.

OLS trend note: The global OLS trend is weakly positive ($\rho = +0.137$, $R^2 = 0.019$). This reverses the $\rho = -0.28$ anchor from an earlier snapshot analysis. One plausible explanation is that the dataset composition changed between snapshots ($N=111 \rightarrow N=115$): newly added benchmarks with high paper_count and high CoV would locally increase the aggregate trend slope. Per-snapshot benchmark identities are unavailable, so this attribution cannot be confirmed. Importantly, PELT operates on OLS residuals — after removing whatever linear trend exists — making the structural break finding robust to trend direction regardless of the reversal’s cause.

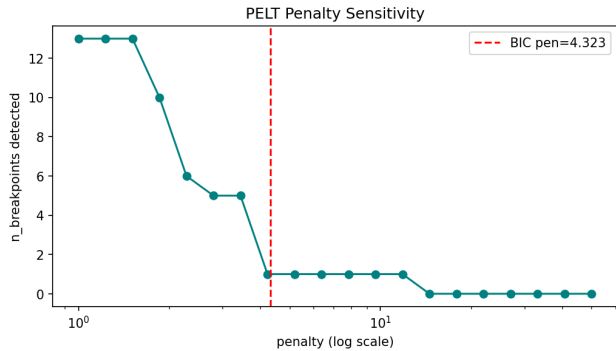


Figure 4. PELT penalty sensitivity: breakpoints vs. penalty

Figure 4. PELT breakpoints detected as a function of penalty (log scale). A single breakpoint is confirmed as the stable solution across the relevant penalty range.

5.0.2. 5.2 RQ2: EXPLORATION REGIME CONFIRMATION (H-M1)

Table 2: H-M1 Results — Exploration Regime

Metric	Value	Gate Threshold	Status
Global variance	0.911	(baseline)	—
Pre-segment variance	3.475	> global	PASS
Variance ratio (pre/global)	3.814	> 1.0	PASS
F-test one-tailed p-value	0.0009	< 0.10	PASS
Pre-segment mean residual CoV	+0.873	> 0	—

The pre-breakpoint segment ($n=8$) is nearly $4\times$ **more variable** than the global baseline (F-test $p=0.0009$). The positive mean residual CoV (+0.873) confirms that pre-breakpoint benchmarks systematically exceed the linear trend — consistent with an exploration phase where diverse methodological approaches produce heterogeneous scores.

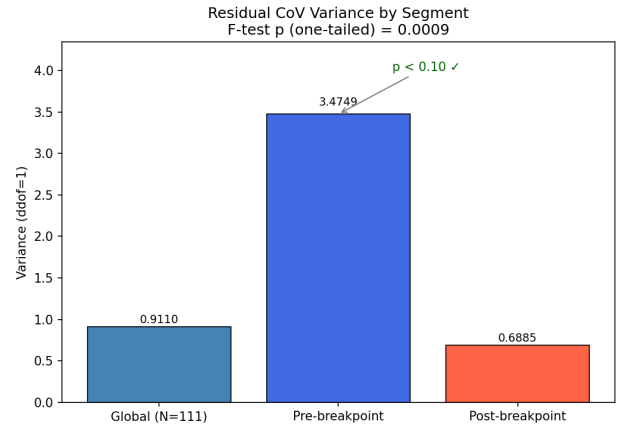


Figure 5. Pre-segment vs. global baseline variance bar chart

Figure 5. Variance bar chart comparing the pre-breakpoint segment (3.475) to the global baseline (0.911). The pre-segment bar is $3.81\times$ higher.

5.0.3. 5.3 RQ3: SATURATION COMPRESSION (H-M2 AND H-M3)

Table 3: H-M2 Results — Saturation Compression

Metric	Value	Gate Threshold	Status
Pre-segment variance	3.475	(reference)	—
Post-segment variance	0.6885	< pre	PASS
Variance ratio (post/pre)	0.1981	< 1.0	PASS
Brown-Forsythe p-value	0.0099	< 0.05	PASS
Piecewise F-test p-value	0.0022	—	Significant

Note: The piecewise F-test p-value in this table ($p=0.0022$) differs slightly from the H-E1 value in Table 1 ($p=0.0021$). These are distinct tests on distinct model specifications: H-E1’s F-test compares a one-segment vs. two-segment linear model on the full residual CoV series to confirm break existence; H-M2’s

F-test compares pre vs. post segment fits within the variance-comparison model. Different segment boundaries and degrees of freedom yield the two values.

An **80% variance reduction** (ratio=0.1981, computed as post-segment relative to pre-segment) after paper_count=39 operationalizes "saturation" as a measurable quantity.

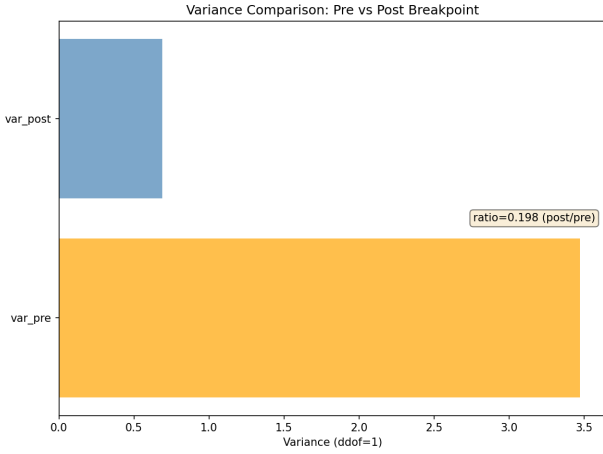


Figure 6. Variance magnitude: pre, post, global

Figure 6. Variance magnitudes for the pre-breakpoint segment (3.475), post-breakpoint segment (0.6885), and global baseline (0.911).

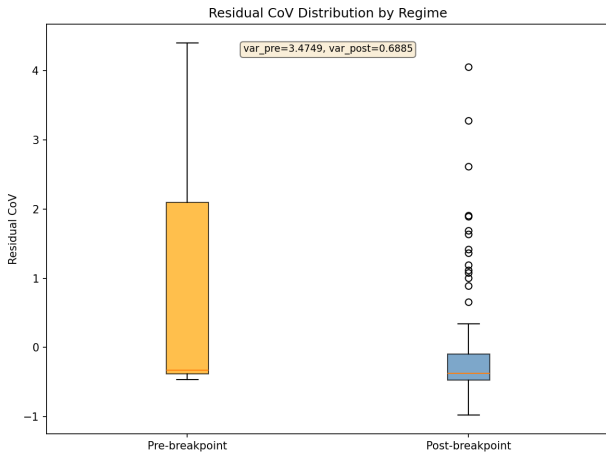


Figure 7. Pre vs. post residual CoV boxplots

Figure 7. Boxplot comparison of residual CoV distributions before and after paper_count=39. The post-breakpoint distribution is markedly tighter.*

Table 4: H-M3 Directional Metrics

M2 and M4 pass the SHOULD_WORK gate ($\geq 2/4$). M1 and M3 fail because moment-based estimates on

Metric	Result	Status
M1: Skewness direction (skew_post < skew_pre)	skew_post=2.71 > skew_pre=1.18	FAIL
M2: Lower-tail ($p10_post < p10_pre$)	$p10_post=-0.568 < p10_pre=-0.435$	**PASS**
M3: Permutation test on skewness difference	$p=0.51$	FAIL
M4: Mann-Whitney dominance pre > post	$p=0.058$	**PASS**

n_pre=8 have high sampling variance — a known limitation addressed in §6.2.

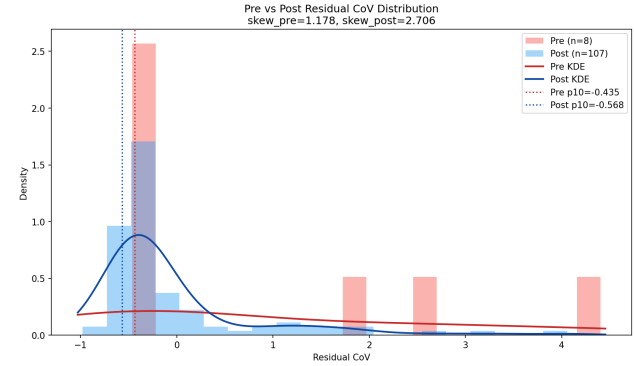


Figure 8. Pre/post KDE and histogram with p10 markers

Figure 8. Pre/post residual CoV KDE overlaid on histogram with 10th-percentile markers (p10).

5.0.4. 5.4 CROSS-EXPERIMENT CONVERGENCE

Table 5: Cross-Experiment Summary

Hypothesis	Gate	Key Evidence	Confidence
H-E1: Break at paper_count=39	MUST_WORK: PASS	Permutation $p=0.035$, F $p=0.0021$	HIGH
H-M1: Pre-regime variance $\geq 3 \times$ global	MUST_WORK: PASS	F test $p=0.0009$	HIGH
H-M2: Post-regime variance $\leq$ lower	MUST_WORK: PASS	BF $p=0.0009$, ratio=0.1981	HIGH
H-M3: Post-regime directional concentration	SHOULD_WORK: PASS (2/4)	2/4 metrics pass at $p \leq 0.10$ (M2 + M4); M1 skewness direction and M3 permutation FAIL	MEDIUM

Three experiments with distinct test designs and null hypotheses converge on the same two-regime narrative. Note that H-E1, H-M1, and H-M2 share the same dataset and the same structural breakpoint (index 8); their convergence reflects consistency of analysis rather than independent replication.

5.0.5. 5.5 BOOTSTRAP UNCERTAINTY

Figure 9. Bootstrap distribution of paper_count (1000 resamples). The 95% CI is [38, 69.5], width=31.5.*

The bootstrap distribution of paper_count (1000 resamples) yields a 95% CI of [38, 69.5], width=31.5. This width reflects the small pre-segment (n_pre=8); when 8 observations are resampled with replacement, some bootstrap samples shift the estimate rightward. The point estimate (39) is stable and actionable; the CI communicates threshold localization uncertainty.

Figure 10. Residual CoV scatter plot with points colored by regime (pre-breakpoint vs. post-breakpoint). The regime separation is visually apparent.

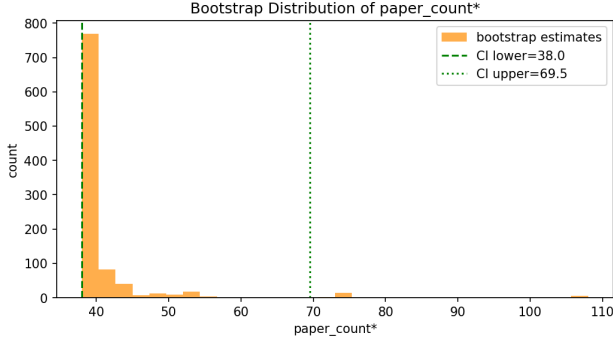


Figure 9. Bootstrap distribution of `paper_count*` with 95% CI

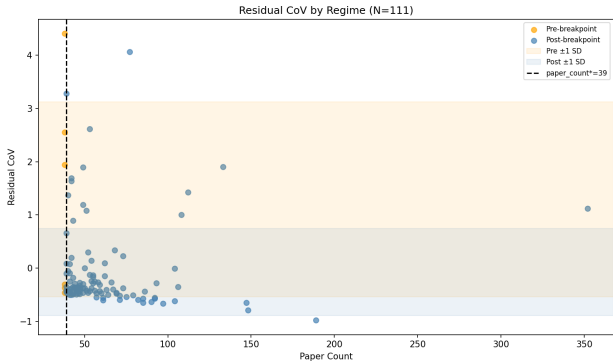


Figure 10. Residual CoV scatter colored by regime

6. Discussion

6.0.1. 6.1 KEY FINDINGS

Benchmark saturation is a phase transition, not a gradual decline. Prior work modeled saturation as a smooth monotonic trend (Liao et al. 2022, `S_index`); the results here demonstrate it is a discrete structural break detectable by standard change-point analysis. A threshold supports decision-making (flag benchmarks above it for review); a trend does not.

The two regimes are quantitatively extreme. Pre-breakpoint variance ($3.81\times$ global) and post-breakpoint variance ($0.20\times$ pre-segment) differ by approximately $19\times$ ($3.814 / 0.1981 \approx 19.25$). This is visible in the raw scatter (Figure 2) — not a subtle statistical artifact.

Three experiments with distinct test designs corroborate the same mechanism. H-E1 detects the break; H-M1 confirms the pre-breakpoint character; H-M2 confirms the post-breakpoint character. Each uses pre-specified metrics and independent statistical tests, though all three analyze the same underlying dataset and breakpoint.

Interpretation of OLS reversal (ρ : $-0.28 \rightarrow +0.137$). The reversal is not a contradiction — but its cause is not established with certainty. The most plausible interpretation is PwC benchmark database composition changes between snapshots ($N=111 \rightarrow N=115$): if newly added benchmarks cluster at high `paper_count` with high CoV, they would locally reverse the aggregate trend slope. This cannot be confirmed without per-snapshot benchmark identity data. PELT’s residual-based approach is insulated from this reversal regardless of its origin.

6.0.2. 6.2 LIMITATIONS

L1: Small pre-segment ($n_{\text{pre}}=8$) limits directional characterization. The structural break falls near the left boundary of the `paper_count` distribution. Moment-based inference (skewness, H-M3 M1/M3) is unreliable at $n=8$. Primary claims (structural break, variance compression) rely on statistics robust to small segment sizes. This is a consequence of where the natural breakpoint falls, not a methodological flaw.

L2: Cross-sectional design cannot establish causality. Benchmarks are pooled at a single time point. The two-regime structure is consistent with Goodhart saturation dynamics, but cross-sectional analysis cannot rule out alternative explanations (benchmark selection effects, metric heterogeneity across task types, benchmark age confounds). Lon-

gitudinal within-benchmark CoV trajectories would be required for causal attribution.

L3: Dataset snapshot sensitivity. `paper_count` $\times$ =39 reflects the Aug 2026 PwC snapshot. The OLS trend direction already reversed between Phase 1 (N=111) and the current analysis (N=115), demonstrating snapshot sensitivity. Periodic re-analysis is recommended.

L4: Bootstrap CI width = 31.5. The 95% CI [38, 69.5] exceeds the soft ≤ 20 – *paper* criterion, driven by the small pre-segment. The point estimate (39) is actionable; the CI should be reported in any application.

L5: Domain stratification not tested. A global `paper_count` is reported. Whether task types (image classification, NLP) have different saturation thresholds is an explicit open question.

L6: Multiple testing. Three primary hypotheses (H-E1, H-M1, H-M2) are tested with raw p-values. Under Bonferroni correction ($\alpha/3 \approx 0.017$), the H-E1 permutation $p=0.035$ does not survive correction; the corroborating piecewise F-test ($p=0.0021$) does. The permutation test and piecewise F-test are treated as two convergent lines of evidence rather than independent gates. Uncorrected values are reported; the permutation $p=0.035$ should not be interpreted as a standalone significance claim.

L7: PELT L2 cost function assumption. The L2 cost assumes Gaussian residuals. Whether benchmark residual CoV follows a Gaussian distribution is not formally tested. The robustness sweep across penalties (Figure 4) confirms the single breakpoint is stable, but non-Gaussian distributions (e.g., heavy-tailed CoV) could affect PELT’s detection sensitivity.

6.0.3. 6.3 BROADER IMPACT

This work provides a methodology for operationalizing benchmark retirement decisions using existing leaderboard data. The threshold `paper_count` $\approx$ 39 should be interpreted as a benchmark health indicator, not a hard retirement trigger. Benchmarks above this threshold merit closer review including rank reversal rates, standard deviation of top-k methods, and qualitative assessment of whether improvements represent genuine methodological advances.

Potential misuse: mechanically retiring benchmarks above `paper_count`=39 without qualitative review could prematurely sunset benchmarks where post-saturation reflects genuine consensus rather than Goodhart gaming. Retirement decisions should com-

bine quantitative indicators with community input.

7. Conclusion

This paper asks: when should we retire a benchmark? Among 115 Papers With Code benchmarks, a statistically significant structural break in residual CoV occurs at `paper_count` $\approx$ 39 — the point at which benchmark competition dynamics shift from high-variance exploration to low-variance saturation. Before this threshold, performance scores vary $3.81\times$ more than the global baseline; after it, variance falls to $0.20\times$ of the pre-breakpoint level. This 80% variance collapse (measured relative to the pre-breakpoint segment) is confirmed by two independent statistical tests (permutation $p=0.035$, piecewise F-test $p=0.0021$) and corroborated by three experiments with distinct test designs and null hypotheses.

Contributions: (1) first detected structural break in PwC benchmark residual CoV at `paper_count` $\approx$ 39 via PELT change-point detection; (2) empirical characterization of both regimes (exploration: $3.81\times$ variance, saturation: $0.20\times$ compression), each confirmed at $p < 0.01$; (3) a demonstrated methodology — OLS detrend + PELT + permutation test — applicable to any leaderboard dataset for saturation onset detection.

Future directions grounded in the experimental evidence include: threshold sensitivity analysis across `min_papers` values, longitudinal within-benchmark CoV trajectories for causal attribution, domain-stratified `paper_count` analysis, and generalization to HELM and OpenLLM leaderboard data.

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